BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

Functional organization and nervous tissue

The written notes for the nervous tissue lecture: the divisions of the nervous system, the parts of a neuron, the chemical synapse, neuron classification, the six glial cells, and myelination and regeneration. Module 5 competencies 173 to 178, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 173. Nervous system divisions

In your course notes: Module 5 packet, Chapter 7, competency 173 (page 55).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Nervous system vs endocrine, Sensory and motor functions, Central nervous system roles, and Reflexes and the feedback loop.

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

The nervous system is the body's fast control system. It senses changes, decides what they mean, and sends out commands. Anatomists divide it in two ways: by where the structures sit (central or peripheral), and by what the motor neurons control (skeletal muscle, the organs, or the gut). You need both divisions, and you need to know which branch sits under which.

By location: central and peripheral

Central nervous system (CNS)

The brain and the spinal cord. The brain sits in the cranial cavity. The spinal cord runs down the vertebral canal and links the brain to the body. The CNS is where information is processed and where commands are issued.

Peripheral nervous system (PNS)

Every neural structure outside the brain and spinal cord: the nerves, the ganglia, and the sensory receptors. There are 12 pairs of cranial nerves, which attach to the brain, and 31 pairs of spinal nerves, which attach to the spinal cord. The PNS carries information between the CNS and every other part of the body.

Structure	What it is
Central nervous system (CNS)	The brain and the spinal cord; it processes information and issues commands
Brain	The CNS organ in the cranial cavity
Spinal cord	The CNS cord in the vertebral canal that links the brain to the body
Peripheral nervous system (PNS)	Every neural structure outside the brain and spinal cord
Nerves	Bundles of axons in the PNS; 12 pairs of cranial nerves and 31 pairs of spinal nerves
Ganglia	Small clusters of neuron cell bodies outside the CNS
Sensory receptors	Structures that detect changes inside and outside the body

By direction: sensory and motor

The PNS has a sensory side and a motor side. The sensory (afferent) division carries signals from receptors toward the CNS. The motor (efferent) division carries commands away from the CNS to the effectors, the muscles and glands that respond.

The motor divisions, by the effector they control

The motor side is split by what it runs. The somatic nervous system sends commands to skeletal muscle, and you control it voluntarily. The autonomic nervous system (ANS) sends commands to smooth muscle, cardiac muscle, and glands, and it works without conscious control. The ANS has two divisions: the sympathetic division for fight or flight (exercise and emergencies), and the parasympathetic division for rest and digest (routine activities). Many organs receive both. The enteric nervous system (ENS) is a network of roughly 100 million neurons in the wall of the digestive tract. It controls the smooth muscle and glands of the gut and can run on its own, though the ANS connects to it and adjusts it. Some textbooks count it as part of the ANS.

Division	Effector it controls	Control
Somatic nervous system	Skeletal muscle	Voluntary
Autonomic nervous system (ANS)	Smooth muscle, cardiac muscle, and glands	Involuntary
Sympathetic division (ANS)	The same ANS effectors, for exercise and emergencies (fight or flight)	Involuntary
Parasympathetic division (ANS)	The same ANS effectors, for routine activities (rest and digest)	Involuntary
Enteric nervous system (ENS)	Smooth muscle and glands in the wall of the digestive tract	Involuntary; can work on its own

The outline, top to bottom

1. Nervous system.
2. Central nervous system (CNS): brain and spinal cord.
3. Peripheral nervous system (PNS): nerves, ganglia, sensory receptors.
4. Sensory (afferent) division of the PNS: toward the CNS.
5. Motor (efferent) division of the PNS: away from the CNS.
6. Somatic nervous system: skeletal muscle.
7. Autonomic nervous system: smooth muscle, cardiac muscle, glands, with sympathetic and parasympathetic divisions.
8. Enteric nervous system: the wall of the digestive tract.

- Sympathetic and parasympathetic belong under autonomic, not under somatic.
- The heart is muscle, but it is cardiac muscle, so it is autonomic, not somatic. Only skeletal muscle is somatic.

- A loop of gut wall has its own neurons, and those belong to the enteric division.

The three functions

Every task the nervous system performs runs through the same three steps.

1. Sensory function: receptors detect a change and send the input toward the CNS.
2. Integrative function: the CNS analyzes the input, stores some of it, and decides on a response.
3. Motor function: the CNS sends commands out to the effectors, which carry out the response.

- Memory tip: Afferent arrives at the CNS. Efferent exits the CNS.
- The motor divisions are named for what they run: somatic for skeletal muscle, autonomic for the organs, enteric for the gut.

Competency 174. Neuron structure

In your course notes: Module 5 packet, Chapter 7, competency 174 (page 57).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Neuron parts and roles, and Neuron structure and signaling.

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

A neuron is the nerve cell that carries signals. Mature neurons are excitable, they can live for a whole lifetime, and they generally do not divide. Every neuron has a cell body and two kinds of processes: dendrites and an axon. The multipolar neuron on the lab slide and the neuron model has many dendrites and one axon, so that is the shape to learn these parts on.

The cell body

The cell body (also called the soma or perikaryon) holds the nucleus, which has a large, dark nucleolus, and most of the cytoplasm and organelles. It is the metabolic center of the neuron and makes the proteins the whole cell needs. Scattered through its cytoplasm are Nissl bodies, clusters of rough endoplasmic reticulum and free ribosomes where those proteins are made. Nissl bodies stain dark on a slide and appear in the cell body and the dendrites, never in the axon hillock or the axon. Neurofibrils, bundles of cytoskeleton filaments, run through the cell and give it its shape. Because the axon depends on the cell body for its proteins, the part of an axon cut away from its cell body dies.

Dendrites, hillock, axon, terminals

Dendrites

The short, heavily branched processes that receive incoming signals. They carry those signals toward the cell body.

Axon hillock

The cone-shaped region where the cell body narrows into the axon. It has no Nissl bodies, so it stains paler than the rest of the cell body. That pale cone is how you find the axon on a slide and tell it apart from a dendrite. Nerve impulses begin at the trigger zone, where the axon hillock meets the first part of the axon.

Axon

The single long process that carries the impulse away from the cell body. Its cytoplasm is the axoplasm and its plasma membrane is the axolemma. An axon can give off side branches called axon collaterals.

Axon terminals

At its far end the axon divides into many fine branches, the axon terminals. The tip of each terminal swells into a synaptic end bulb that contacts the next cell.

The direction a signal travels

1. Dendrites receive the incoming signal and carry it toward the cell body.
2. The cell body collects the signals arriving from the dendrites.
3. The axon hillock is where the impulse starts, at the trigger zone where the hillock joins the axon.
4. The axon carries the impulse away from the cell body.
5. The axon terminals branch at the end of the axon and end in synaptic end bulbs that pass the signal to the next cell.

Structure	What it is
Cell body	The soma or perikaryon; it holds the nucleus and cytoplasm and is the metabolic center of the neuron
Nucleus and nucleolus	The cell's nucleus, with a large, dark-staining nucleolus, inside the cell body
Nissl bodies	Clusters of rough ER and ribosomes in the cell body and dendrites; the site of protein synthesis
Neurofibrils	Bundles of cytoskeleton filaments that give the neuron its shape and support
Microtubules	Cytoskeletal tubes that transport materials along the neuron
Dendrites	The branched processes that receive incoming signals and carry them toward the cell body
Axon hillock	The cone-shaped region where the cell body joins the axon; it has no Nissl bodies
Axon	The single long process that carries the impulse away from the cell body
Axoplasm and axolemma	The cytoplasm of the axon and the plasma membrane of the axon
Axon terminals	The fine branches at the far end of the axon that end in synaptic end bulbs
Lipofuscin	A yellow-brown pigment that builds up in neurons with age; it does not appear to harm the cell

- Memory tip: Dendrites deliver to the cell body. The axon carries the signal away.
- On a slide, the pale cone with no Nissl bodies marks the start of the axon.
- The order is dendrites, cell body, axon hillock, axon, axon terminals. The hillock comes between the cell body and the axon, never after the axon.

Competency 175. Synapse structures

In your course notes: Module 5 packet, Chapter 7, competency 175 (page 58).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Synapses and chemical messengers.

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

A synapse is the junction where a neuron passes its signal to another cell. The cell that sends the signal is the presynaptic neuron. The cell that receives it is the postsynaptic cell, which can be another neuron, a muscle fiber, or a gland cell. The synapse between a motor neuron and a skeletal muscle fiber is the neuromuscular junction.

The parts of a chemical synapse

At a chemical synapse the two cells do not touch. A narrow gap, the synaptic cleft, separates them. It is only about 20 to 50 nanometers wide. The signal crosses this gap as a chemical, a neurotransmitter. The other kind of synapse, an electrical synapse, joins two cells directly through gap junctions and is much less common.

On the sending side, each axon terminal ends in a swelling called a synaptic end bulb. Your lab list calls it the synaptic bulb (terminal end). Inside the end bulb are many synaptic vesicles, tiny membrane sacs that store neurotransmitter. The membrane of the end bulb that faces the cleft is the presynaptic membrane. Across the cleft, the postsynaptic membrane holds the receptors that the neurotransmitter binds to. Only the presynaptic side has vesicles, so a chemical synapse passes its signal in one direction only.

How the signal crosses, in order

1. Synaptic end bulb: the nerve impulse arrives at the end of the axon terminal.
2. Synaptic vesicles fuse with the presynaptic membrane and release neurotransmitter.
3. Synaptic cleft: the neurotransmitter diffuses across the gap.
4. Postsynaptic membrane: the neurotransmitter binds receptors, and the signal passes to the next cell.

Structure	What it is
Synapse	The site where a neuron communicates with another neuron, a muscle cell, or a gland
Chemical synapse	A synapse with no direct contact; the signal crosses as neurotransmitter molecules
Synaptic end bulbs	The swellings at the tips of the axon terminals that hold the synaptic vesicles
Synaptic vesicles	The tiny sacs inside the end bulbs that store neurotransmitter
Synaptic cleft	The narrow gap between the two cells that the neurotransmitter diffuses across
Postsynaptic membrane	The membrane of the receiving cell, which holds the neurotransmitter receptors
Neurotransmitters	The chemical messengers that carry the signal across the cleft to the next cell

Three synapse types, named for the site the axon contacts

Each name starts with "axo" for the presynaptic axon and ends with the part of the next neuron it touches.

Synapse type	Where the axon lands	Notes
Axodendritic synapse	On a dendrite	The most common type
Axosomatic synapse	On a cell body (soma)	
Axoaxonic synapse	On another axon	

- Memory tip: Read the synapse names left to right. The first half is always the axon, and the second half is where it lands: dendrite, soma, or axon.
- There is a gap between the two cells. If your end bulb touches the next cell in a drawing, open up the cleft.
- Vesicles belong in the sending end bulb, not in the receiving cell.

Competency 176. Neuron classification

In your course notes: Module 5 packet, Chapter 7, competency 176 (page 59).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Neuron structure and signaling.

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

Neurons are classified in two ways. The structural classification counts how many processes leave the cell body. The functional classification asks which direction the neuron carries its signal, toward the CNS or away from it. One neuron gets a name from each system, so a touch neuron is both unipolar and sensory.

Structural classification, by number of processes

Multipolar neuron

Many dendrites and one axon. It is the most common type in the body. Motor neurons and interneurons are multipolar, and so is the neuron on your lab slide. Two multipolar neurons you will see on slides are the pyramidal cells of the cerebral cortex and the Purkinje cells of the cerebellum.

Bipolar neuron

One dendrite and one axon, leaving from opposite ends of the cell body. Bipolar neurons are rare. They are found in special sense organs: the retina of the eye, the olfactory epithelium of the nose, and the inner ear.

Unipolar neuron

One short process that leaves the cell body and then splits like the letter T into two branches. Some textbooks call it pseudounipolar. The peripheral branch starts at a receptor, and the central branch enters the CNS. Most sensory neurons are unipolar, and their cell bodies sit in the dorsal root ganglia and the cranial nerve ganglia.

Type	Processes	Example
Multipolar neuron	Many dendrites and one axon; the most common type	A motor neuron in the anterior horn of the spinal cord; pyramidal cells; Purkinje cells
Bipolar neuron	One dendrite and one axon	Neurons in the retina, the olfactory epithelium, and the inner ear
Unipolar neuron	One process that splits into a peripheral branch and a central branch	A touch or pain sensory neuron with its cell body in a dorsal root ganglion
Pyramidal cells	A large multipolar neuron with a pyramid-shaped cell body	The cerebral cortex
Purkinje cells	A large multipolar neuron with a fan of dendrites	The cerebellar cortex

Functional classification, by direction

Sensory (afferent) neurons

Carry signals from receptors toward the CNS. A neuron that carries touch from the skin of a fingertip to the spinal cord is a sensory neuron. Most are unipolar.

Motor (efferent) neurons

Carry commands away from the CNS to an effector. A motor neuron whose cell body sits in the anterior horn of the spinal cord and whose axon runs to the biceps brachii is an example. Motor neurons are multipolar.

Interneurons

Also called association neurons. They lie entirely within the CNS and connect sensory neurons to motor neurons and to each other. The neurons in the spinal cord gray matter that link the sensory and motor neurons of a withdrawal reflex are interneurons. Most neurons in the body are interneurons, and they are multipolar.

Type	Direction	Example
Sensory (afferent) neurons	Toward the CNS; most are unipolar	A touch neuron from the fingertip to the spinal cord
Motor (efferent) neurons	Away from the CNS; multipolar	An anterior horn motor neuron to the biceps brachii
Interneurons	Within the CNS, between other neurons; multipolar	Spinal cord neurons linking the sensory and motor neurons of a withdrawal reflex

- Memory tip: SAME. Sensory is Afferent, Motor is Efferent.
- For shape, count the processes leaving the cell body: many, two, or one. Bipolar has two and unipolar has one.
- In a skin-to-cord-to-muscle pathway, the sensory neuron is the unipolar one. The interneuron inside the cord and the motor neuron out to the muscle are both multipolar.

Competency 177. Neuroglia identification

In your course notes: Module 5 packet, Chapter 7, competency 177 (page 60).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Neuroglia: supporting cells, and Glial types, myelin, and protection.

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

Neuroglia, or glial cells, are the support cells of nervous tissue. They are smaller than neurons and very numerous, though textbooks differ on how many glia there are for each neuron. Unlike mature neurons, they can still divide. In the CNS they make up about half the volume of the tissue. On a slide, the many small dark nuclei around the large neuron cell bodies belong to glia. There are six types. Four are in the CNS and two are in the PNS.

The four glia of the CNS

Astrocytes

Star-shaped cells and the most numerous glia in the CNS. Their processes wrap around neurons and capillaries. They hold neurons in place, help form the blood-brain barrier, and keep the fluid around neurons at the right chemical balance. In the embryo they guide growing neurons, and after an injury they fill the damaged area with scar tissue.

Oligodendrocytes

Cells with a few long processes. Each process wraps around a segment of an axon to form myelin, so one oligodendrocyte myelinates parts of several different axons. They are the myelin-forming cells of the CNS.

Microglia

Small cells that act as phagocytes. They engulf microbes and clear away debris from dead and damaged cells. They are the CNS's own immune defense.

Ependymal cells

Cuboidal to columnar cells, many with cilia, that line the ventricles of the brain and the central canal of the spinal cord. In the choroid plexuses they help produce cerebrospinal fluid, and their cilia help it circulate.

Glial cell	What it looks like	Support role
Astrocytes	Star-shaped; the most numerous CNS glia	Support neurons, help form the blood-brain barrier, keep the chemical environment around neurons stable, guide neuron growth, form scar tissue after injury
Oligodendrocytes	A few long processes that wrap axons	Form the myelin sheaths of the CNS, each cell on several axons
Microglia	Small cells	Phagocytes that remove microbes and debris
Ependymal cells	Cuboidal to columnar, many with cilia	Line the ventricles and central canal; help produce and circulate cerebrospinal fluid

The two glia of the PNS

Schwann cells

Wrap around PNS axons. Each Schwann cell forms the myelin around one segment of a single axon. Schwann cells also enclose unmyelinated PNS axons, and they guide a damaged axon as it regrows.

Satellite cells

Flat cells that surround the cell bodies of neurons in ganglia. They support those cell bodies and regulate the exchange of materials between the cell bodies and the fluid around them.

Glial cell	What it looks like	Support role
Schwann cells	Wrapped around a segment of one axon	Form the myelin sheath around PNS axons; aid axon regeneration
Satellite cells	Flat cells around neuron cell bodies	Support the cell bodies of PNS neurons in ganglia and regulate their surroundings

- Memory tip: The two PNS glia both start with S, Schwann and Satellite. Every other glial cell is in the CNS.
- Two cells make myelin: oligodendrocytes in the CNS and Schwann cells in the PNS. One cell helps form the blood-brain barrier: the astrocyte.
- If you sort the six cells into a CNS side and a PNS side, the CNS side should have four and the PNS side two.

Competency 178. Myelination and regeneration

In your course notes: Module 5 packet, Chapter 7, competency 178 (page 62).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Glial types, myelin, and protection, and Nerves, tracts, white and gray matter.

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

The myelin sheath and its parts

The myelin sheath is a covering made of many layers of a glial cell's plasma membrane, wrapped tightly around an axon. Because it is mostly lipid with some protein, it looks white. It insulates the axon and makes the nerve impulse travel much faster. An axon with this covering is a myelinated axon.

Myelin is laid down in segments along the axon. The short gaps between neighboring segments are the nodes of Ranvier, where the axolemma is left bare. The myelin-covered stretch between two nodes is an internode. In a myelinated axon the impulse jumps from node to node. This fast form of conduction is called saltatory conduction.

The neurolemma is the outermost layer of a Schwann cell. It holds the Schwann cell's nucleus and most of its cytoplasm, and it sits outside the myelin layers. Only Schwann cells have one, so the neurolemma is found only around PNS axons.

Structure	What it is
Myelin sheath	A multilayered covering of lipid and protein wrapped around an axon; it insulates the axon and speeds up the impulse
Nodes of Ranvier	The gaps in the myelin sheath, spaced along the axon
Internode	The myelin-covered segment between two nodes
Saltatory conduction	The fast conduction in which the impulse jumps from node to node
Neurolemma	The outer layer of a Schwann cell, holding its nucleus and cytoplasm; PNS only

PNS and CNS myelination compared

In the PNS, the myelin is made by Schwann cells. Each Schwann cell wraps one segment of one axon, so a long axon needs a row of Schwann cells along its length, with a node of Ranvier between neighboring cells. In the CNS, the myelin is made by oligodendrocytes. One oligodendrocyte sends processes to several axons and myelinates a segment of each. Oligodendrocytes have no neurolemma.

That difference decides whether a cut axon can regrow. When a PNS axon is cut, the part beyond the cut breaks down. If the cell body is intact, the Schwann cells line up and their neurolemma forms a regeneration tube that guides the regrowing axon back toward its target. In the CNS there is no neurolemma to guide regrowth.

Oligodendrocytes also release substances that block regrowth, and astrocytes fill the area with scar tissue. Regeneration in the CNS is very limited.

Feature	PNS myelination	CNS myelination
Cell that makes the myelin	Schwann cell	Oligodendrocyte
Axons per glial cell	One segment of one axon	Segments of several axons
Neurolemma	Present; the outer layer of the Schwann cell	Absent
Axon regeneration	Possible if the cell body survives; the neurolemma forms a regeneration tube that guides the axon	Very limited; no neurolemma, oligodendrocytes block regrowth, and astrocytes form scar tissue

- Memory tip: Schwann cells serve one axon segment each and leave a neurolemma. Oligodendrocytes reach several axons and leave none. That is why a cut nerve in the arm can regrow and a cut tract in the spinal cord usually cannot.
- If your drawing shows one Schwann cell reaching several axons, you drew an oligodendrocyte.
- Myelination increases from birth to maturity, so an infant's nerve conduction is slower than an adult's.

What this means for patients

- Mature neurons generally do not divide, so tumors of nervous tissue rarely come from neurons. Most come from the glia, which can still divide. A glioma is a tumor of glial cells, and it is the most common primary malignant brain tumor.
- Myelin matters as much as the axon it wraps. In multiple sclerosis, the immune system destroys myelin around CNS axons, and signals slow down or fail.
- The presence of the neurolemma on Schwann cells, and its absence on oligodendrocytes, is why a cut nerve in the arm can regrow while damage in the brain or spinal cord usually cannot be repaired.
- The cell body keeps the whole neuron alive. The axon depends on the cell body for its proteins, so the part of an axon cut off from its cell body dies. An injury close to the cell body is more likely to kill the whole neuron than an injury far out along the axon.

Worked questions

Eight questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 173

The motor neurons that make the heart beat faster during exercise belong to which division?

A. Somatic nervous system

B. Enteric nervous system

C. Autonomic nervous system

D. Sensory division of the PNS

Answer: C. The heart is cardiac muscle, and cardiac muscle, smooth muscle, and glands are the effectors of the autonomic nervous system. Speeding the heart during exercise is the job of its sympathetic division.

Why the others are wrong

- **A** The somatic nervous system controls only skeletal muscle, and it is voluntary. The heart is muscle, but it is cardiac muscle, not skeletal.
- **B** The enteric nervous system is the network of neurons in the wall of the digestive tract. It runs the gut, not the heart.
- **D** The sensory division carries signals toward the CNS. Commands to an effector travel on the motor side.

QUESTION 2 · COMPETENCY 173

How many pairs of spinal nerves attach to the spinal cord?

A. 31

B. 12

C. 24

D. 43

Answer: A. There are 31 pairs of spinal nerves attached to the spinal cord and 12 pairs of cranial nerves attached to the brain. Both sets belong to the PNS.

Why the others are wrong

- **B** Twelve is the number of pairs of cranial nerves, which attach to the brain.
- **C** Twenty-four pairs is not a count from this chapter. The spinal cord gives off 31 pairs.
- **D** Forty-three is the total if you add the 12 cranial pairs and the 31 spinal pairs together. The question asks only about the spinal nerves.

QUESTION 3 · COMPETENCY 174

On a slide, the pale cone-shaped region where the cell body narrows into the axon has no Nissl bodies. What is it?

A. Dendrite

B. Synaptic end bulb

C. Nucleolus

D. Axon hillock

Answer: D. The axon hillock is the cone where the cell body joins the axon. It has no Nissl bodies, so it stains paler than the rest of the cell body, and the impulse begins at the trigger zone where the hillock meets the axon.

Why the others are wrong

- **A** Dendrites are the short branched processes that receive signals, and they do contain Nissl bodies, so they stain dark like the cell body.
- **B** The synaptic end bulb is the swelling at the tip of an axon terminal, at the far end of the axon, not where the axon begins.
- **C** The nucleolus is the large, dark-staining body inside the nucleus. It is dark, not pale, and it sits in the cell body.

QUESTION 4 · COMPETENCY 174

Which list gives the order a signal travels through a neuron?

A. Axon, axon hillock, cell body, dendrites, axon terminals

B. Dendrites, cell body, axon hillock, axon, axon terminals

C. Cell body, dendrites, axon, axon hillock, axon terminals

D. Axon terminals, axon, axon hillock, cell body, dendrites

Answer: B. Dendrites receive the signal and carry it to the cell body. The impulse starts at the axon hillock, travels down the axon, and reaches the axon terminals, where the synaptic end bulbs pass it to the next cell.

Why the others are wrong

- **A** This runs the first four parts backward. The axon carries the signal away from the cell body, not into it.
- **C** Dendrites come before the cell body, and the axon hillock comes before the axon, not after it. The hillock is where the cell body narrows into the axon.
- **D** This is the whole path in reverse. The axon terminals are the end of the line, where the signal leaves the neuron.

QUESTION 5 · COMPETENCY 175

Where are the synaptic vesicles found at a chemical synapse?

A. Inside the synaptic end bulb of the presynaptic neuron

B. In the synaptic cleft

C. In the postsynaptic membrane

D. In the dendrites of the receiving cell

Answer: A. Synaptic vesicles are tiny membrane sacs that store neurotransmitter inside the synaptic end bulb. Only the presynaptic side has vesicles, which is why a chemical synapse passes its signal in one direction only.

Why the others are wrong

- **B** The synaptic cleft is the 20 to 50 nanometer gap between the two cells. Neurotransmitter diffuses across it; the vesicles stay in the end bulb.
- **C** The postsynaptic membrane holds the receptors that the neurotransmitter binds to. It has no vesicles.
- **D** Dendrites are on the receiving side of the most common synapse, the axodendritic synapse. Vesicles belong to the sending axon, not the receiving dendrite.

QUESTION 6 · COMPETENCY 176

A neuron with one short process that leaves the cell body and splits like a T, with its cell body in a dorsal root ganglion, is

A. multipolar and a motor neuron

B. bipolar and found in the retina

C. multipolar and an interneuron

D. unipolar and a sensory neuron

Answer: D. One process that splits into a peripheral branch from a receptor and a central branch into the CNS is the unipolar shape. Most sensory neurons are unipolar, and their cell bodies sit in the dorsal root ganglia and cranial nerve ganglia.

Why the others are wrong

- **A** Motor neurons are multipolar, with many dendrites and one axon, and their cell bodies sit inside the CNS, for example in the anterior horn of the spinal cord.
- **B** A bipolar neuron has one dendrite and one axon leaving opposite ends of the cell body, two processes, not one. Bipolar neurons are found in the retina, the olfactory epithelium, and the inner ear.
- **C** Interneurons are multipolar and lie entirely within the CNS. A cell body in a ganglion is outside the CNS.

QUESTION 7 · COMPETENCY 177

Which glial cell lines the ventricles of the brain and the central canal of the spinal cord and helps produce cerebrospinal fluid?

A. Astrocyte

B. Microglial cell

C. Ependymal cell

D. Satellite cell

Answer: C. Ependymal cells are cuboidal to columnar cells, many with cilia, that line the ventricles and the central canal. In the choroid plexuses they help produce cerebrospinal fluid, and their cilia help it circulate.

Why the others are wrong

- **A** Astrocytes are the star-shaped cells that support neurons, help form the blood-brain barrier, and keep the chemical environment stable. They do not line the ventricles.
- **B** Microglia are the small phagocytes of the CNS. They remove microbes and debris.
- **D** Satellite cells are PNS glia. They surround and support the cell bodies of neurons in ganglia.

QUESTION 8 · COMPETENCY 178

A nerve in the forearm is cut, but the cell bodies of its neurons survive. Why can these axons regrow when a cut tract in the spinal cord usually cannot?

A. PNS axons have no myelin, so nothing blocks their regrowth

B. The neurolemma of the Schwann cells forms a regeneration tube that guides the axon

C. Oligodendrocytes in the PNS release substances that promote regrowth

D. Astrocytes in the PNS clear the path of scar tissue

Answer: B. When a PNS axon is cut, the Schwann cells line up and their neurolemma, the outer layer holding the Schwann cell's nucleus and cytoplasm, forms a regeneration tube that guides the regrowing axon back toward its target. Oligodendrocytes have no neurolemma, so the CNS has no such tube.

Why the others are wrong

- **A** Many PNS axons are myelinated, by Schwann cells. Myelin itself is not what blocks regrowth; the missing guide is the problem in the CNS.
- **C** Oligodendrocytes are CNS glia, not PNS glia, and they release substances that block regrowth rather than promote it.
- **D** Astrocytes are CNS glia, and they work against regrowth by filling the damaged area with scar tissue. They are not in the PNS.